Termination Analysis of Probabilistic Programs

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Termination of Turing Machines

The halting problem

To determine whether a given program halts for a given input.

The halting problem is undecidable (Turing, 1937).

The universal halting problem

To determine whether a given program halts for all possible inputs.

Strictly more difficult than the halting problem.

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Collatz Conjecture

The "simplest" unsolved math problem:
to check if the program terminates for all $n \in \mathbb{N}$

```
while (n > 1){
    if (n % 2 == 0)
        n := n/2
    else
        n := 3*n+1
}
```

How to prove termination?

Use a **variant function** *on the program's state space whose value — on each loop iteration — is monotonically decreasing with respect to a (strict) well-founded relation.*

Peter Naur. Proof of Algorithms by General Snapshots. 1966

Proof rules in Hoare logic

$$\frac{\{I \wedge G\}C\{I\}}{\{I\}\texttt{while } G \texttt{ do } C\{I \wedge \neg G\}} \qquad \text{(partial correctness)}$$

$$\frac{\{I \wedge G \wedge V = z\}C\{I \wedge V < z\}}{\{I\}\texttt{while } G \texttt{ do } C\{I \wedge \neg G\}} \qquad \text{(total correctness)}$$

- $I : \Sigma \to \{\top, \bot\}$ is the **invariant**.
- $V : \Sigma \to \mathbb{N}$ is the **variant**, also called the **ranking function**.

Probabilistic Programs

What?

- Programs with random assignments, conditioning and usual control-flow constructs

Why?

- for describing randomized algorithms.
- for designing probabilistic models.

Syntax and Semantics

- Probabilistic Guarded Command Language (pCGL) syntax:

$$C \;\hat{=}\; \mathtt{skip} \mid x \leftarrow e \mid x \sim \mu \mid C_1; C_2 \mid C_1[p]C_2 \mid \\ \mathtt{if}(\varphi)\ C_1\ \mathtt{else}\ C_2 \mid \mathtt{while}(\varphi)\ C$$

- $x \sim \mu$: the value of x is sampled from a distribution μ.
- $C_1[p]C_2$: choose C_1 with probability p and choose C_2 with probability $(1-p)$.
- A program state is a distribution!

Termination Analysis

For deterministic programs

- A program either terminates or not on a given input.
- Terminating programs have a finite run-time.
- Having a finite run-time is compositional.

All these facts do not hold for probabilistic programs!

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Certain Termination

```
while (x > 0) {
    x := x-1 [1/2] x:= x-2
}
```

This program terminates for all integer inputs x.

Almost-Sure Termination (AST)

```
x = 1;
while (x > 0) {
    x := 0 [1/2] x:= 1
}
```

This program does not always terminate.
It diverges with probability 0.
It **almost surely terminates** (with probability 1).

AST with finite expected runtime

```
x = 1;
while (x > 0) {
    x := 0 [1/2] x:= 1
}
```

It almost surely terminates in finite expected time.

$$\mathbb{E}[\mathsf{T}] = \sum_{i=1}^{\infty} i \cdot \mathbb{P}(\mathsf{T} = i) = \sum_{i=1}^{\infty} i \cdot (\frac{1}{2})^{i-1} \cdot (1 - \frac{1}{2}) = 2$$

AST with infinite expected runtime

```
x = 1;
while (x > 0) {
    x := x-1 [1/2] x:= x+1
}
```

It almost surely terminates,
but requires an infinite expected time.

Not Almost-Sure Termination

```
function P:
    skip [1/2] {call P; call P; call P}
```

This program terminates with probability $\frac{\sqrt{5}-1}{2}$.

$$x = \frac{1}{2} + \frac{1}{2}x^3$$

Problems

- **Qualitative analysis**: check almost surely termination (AST)
  - Positive AST: ... and in an expected finite number of steps
  - Unbounded AST:... and in an expected infinite number of steps
- **Quantitative analysis**: to compute a lower/upper bound $p \in [0, 1]$ on termination probability.

Martingales

- A stochastic process $X = \{X_0, X_1, X_2, \dots\}$ is a **martingale** if
  - $\{X\}_{n\in\mathbb{N}}$ is adapted to a filtration $\{\mathcal{F}\}_{n\in\mathbb{N}}$;
  - X_n is integrable for each n, i.e., $\mathbb{E}(|X_n|) < \infty$,
  - with probability 1 (i.e., almost surely),

$$\mathbb{E}[X_{n+1} \mid \underbrace{X_0, X_1, \dots, X_n}_{\mathcal{F}_n}] = X_n.$$

- ... is a **supermartingale** if

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n.$$

Ranking Supermartingales

A discrete-time stochastic process $X = \{X_n\}_{n\in\mathbb{N}}$ adapted to a filtration $\{\mathcal{F}_n\}_{n\in\mathbb{N}}$ is a **ranking supermartingale (RSM)**

- (integrability) $\mathbb{E}[|X_n|] < \infty$ for $n \in \mathbb{N}$;
- (lower-bound) $\exists k \leq 0$ s.t. $X_n \geq K$ almost surely for $n \in \mathbb{N}$;
- (decrease) $\exists \epsilon > 0$ s.t. $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n - \epsilon \cdot \mathbf{1}_{X_n \geq 0}$ almost surely.

ranking function: decrease by ϵ at each step.
RSM: expected to decrease by ϵ at each step.

Chakarov and Sankaranarayanan, 2013

RSM for AST

A RSM with a positive initial position
almost surely becomes negative

$\mathbb{P}(Z_X < \infty) = 1$ and $\mathbb{E}[Z_X] \leq \frac{\mathbb{E}[X_0]-K}{\epsilon}$
where Z_X denotes the first time X drops below 0.

RSM Maps

- How to map a loop to a stochastic process?
- For simplicity, consider a discrete-time dynamical system

$$x' = f(x, d), \quad d \sim \mu$$

Prove: starting from X_0, the system will reach a target region X_T.

RSM Maps

An RSM map consists of

- An **invariant** $\Omega \subseteq \mathbb{R}^n$ as an over-approximation of the reachable set.
- A **variant** function $V : \Omega \to \mathbb{R}$

$$
\begin{aligned}
&V(x) \leq 0 && \forall x \in T \\
&0 \leq V(x) \leq H && \forall x \in \Omega \setminus T \\
&\mathbb{E}[V(f(x,d))] \leq V(x) - \epsilon && \forall x \in \Omega \setminus T
\end{aligned}
$$

RSM Maps

Theoretical guarantees?

- **Soundness**

$$\text{an RSM exists} \implies \text{PAST}$$

- **Completeness** (proved for integer-valued programs)

$$\text{PAST} \implies \text{an RSM exists}$$

Synthesis?

- **Template-based**: Assume $V(x)$ is of certain parametric forms, usually a linear or polynomial expression.
- To compute $\mathbb{E}[V(f(x,d))]$, we might need moments of d, i.e. $\mathbb{E}[d], \mathbb{E}[d^2], \ldots$

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AST with infinite expected runtime

```
while ( 0 < x ) {
    x := x-1 [1/2] x:= x+1
}
```

Still beyond reach.

“Walled” programs

```
while ( 0 < x < M ) {
    x := x-1 [1/2] x:= x+1
}
```

After adding a wall, the program becomes AST.

Borel–Cantelli Lemma (0-1 Law)

- Let $E_1, E_2, \ldots$ be a sequence of events in some probability space.
- **First Borel-Cantelli Lemma**

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty \implies \mathbb{P}(\limsup_{n\to\infty} E_n) = 0$$

- **Second Borel-Cantelli Lemma.** If the events E_n are independent

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) = \infty \implies \mathbb{P}(\limsup_{n\to\infty} E_n) = 1$$

Example: the infinite monkey theorem.

Parametric RSM

A non-negative stochastic process $X = \{X_n\}_{n\in\mathbb{N}}$ is a **parametric ranking supermartingale (PRSM)** if

- for each $n \in \mathbb{N}$, $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n$;
- for each $n \in \mathbb{N}$, $\mathbb{P}(X_{n+1} \leq X_n - d(X_n) \mid \mathcal{F}_n) \geq p(X_n)$, for some functions $d : \mathbb{R} \to \mathbb{R}_{\geq 0}$ and $p : \mathbb{R} \to [0, 1]$, both non-increasing and strictly positive on positive reals.

there is a positive probability of a strict decrease.

McIver, et al. 2018

AST with infinite expected runtime

```
while ( 0 < x ) {
    x := x-1 [1/2] x:= x+1
}
```

Let $V(x) := x$, $d(x) := 1$, $p(x) = 1/2$.

$$\mathbb{E}[x] \leq x$$

$$\mathbb{P}(x' \leq x - 1) \geq \frac{1}{2}$$

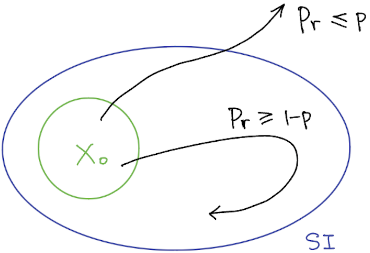
Completeness

- PRSM rule is conjectured to be incomplete.
- Proved to be complete by Majumdar and Sathiyanarayana (POPL'25).
- Complete for discrete-time **weak Feller process** (L-CSS'24)
  - The transition semigroup maps bounded continuous functions to bounded continuous functions.
  - A weak Feller process can be continuous-time.

Probabilistic Termination

- Probabilistic Termination: to find lower/upper bound for termination.
- A **p-stochastic invariant (p-SI)** is a set Ω such that

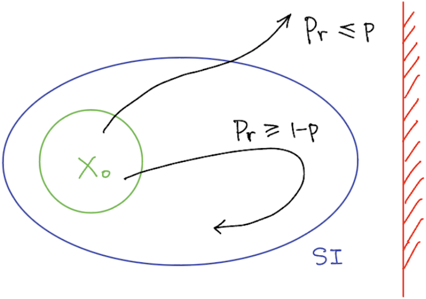
$$\mathbb{P}(\text{a trajectory starting from } X_0 \text{ leaves } \Omega) \leq p$$



Chatterjee22CAV, Xue25arxiv

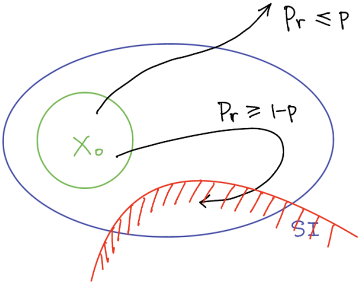
Probabilistic Termination Upper Bound p

Find a p-SI Ω s.t. $\Omega \cap X_T = \emptyset$.



Probabilistic Termination Lower Bound $1 - p$

Find a p-SI Ω
and prove almost surely reach $\Omega \cap X_T$ within Ω.



Summary:

- AST = positive AST + unbounded AST
    - postive AST: **Ranking Supermartingales (RSM)**.
    - unbounded AST: **Parametric RSM**.
- Probabilistic Termination: **Stochastic Invariants**.

Thanks! & Questions?

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